Advanced Generative AI



Benchmark and Evaluation of LLM Capabilities



Quick Recap



- What is the primary purpose of fine-tuning Language Models (LLMs), and how is it achieved?
- Why is hyperparameter adjustment crucial in the fine-tuning process of LLMs, and when are these hyperparameters set?

Engage and Think



Imagine being part of an innovative tech team tasked with revolutionizing user experience by analyzing vast amounts of global user-generated content, reviews, and feedback, while simultaneously developing a cutting-edge virtual assistant to enhance everyday interactions. Your responsibilities include summarizing customer sentiments, overcoming language barriers through advanced translation techniques, and employing chatbots for real-time user engagement. To ensure your virtual assistant comprehends and responds accurately to users, you must excel in benchmarking and evaluating the performance of large language models (LLMs).

What challenges might you encounter, and how would you design a benchmark to assess the LLMs' capabilities?

Learning Objectives

By the end of this lesson, you will be able to:

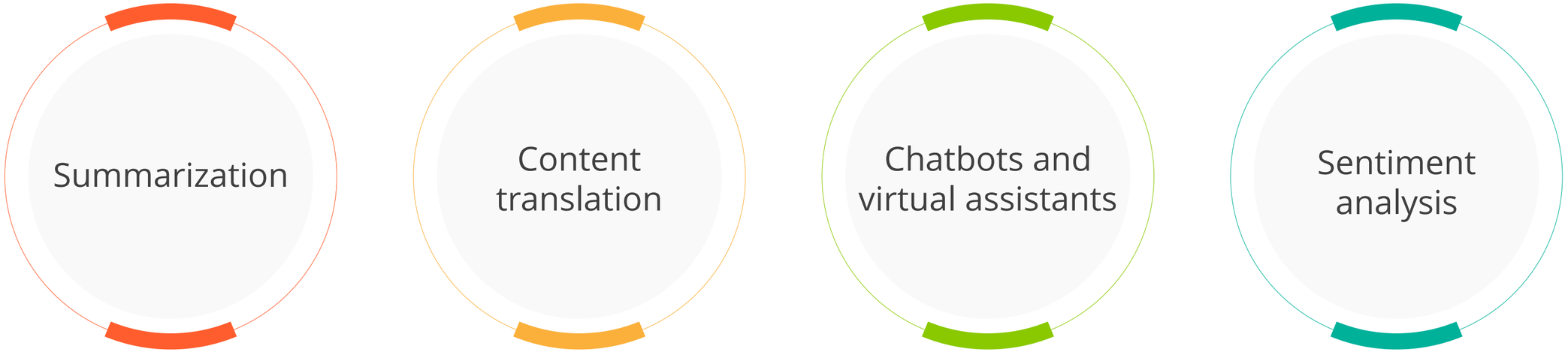
- Apply text summarization techniques for condensing lengthy content using LLM models.
- Build and utilize LLM-based chatbots or virtual assistants for automating tasks or retrieving information.
- Analyze and apply principles from HELM and GLUE benchmarks for large language model evaluation and practical language understanding enhancement.
- Assess SuperGLUE and BIG-bench benchmark results to create new testing scenarios and make informed decisions.



LLM Capabilities

LLM Capabilities

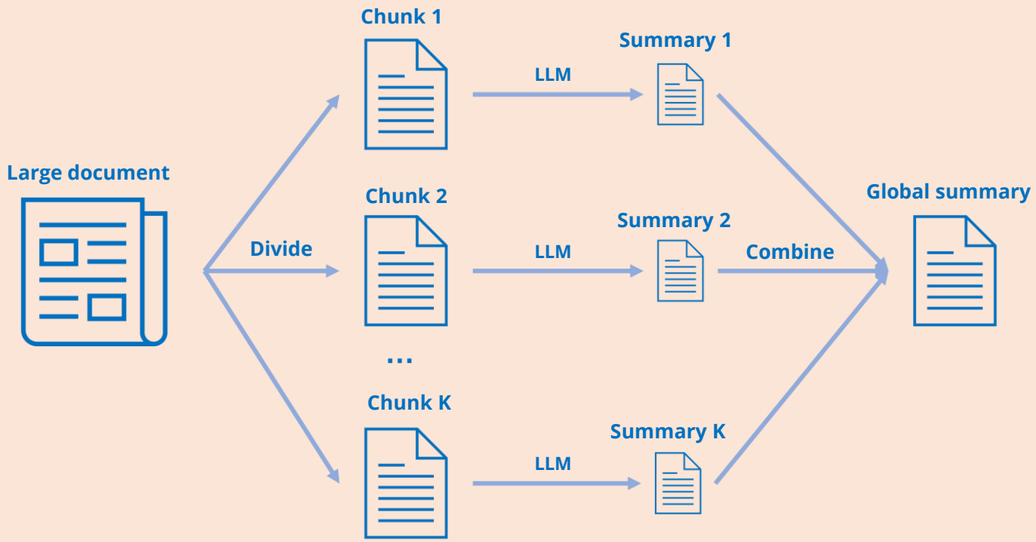
The four major capabilities of LLMs are:



Summarization

Summarization

LLM summarization is a technique that uses LLMs to generate summaries of long texts or documents.



- The **refine** method breaks down a lengthy text into smaller chunks that fit the model's context, summarizes each chunk, concatenates the summaries, and uses the LLM to create a final summary.
- Another method combines individual document summaries from the **Map** step into a cohesive text string.

Summarization

LLM summarization can be used to:

- Control tone and writing style
- Obtain semantically-coherent topics
- Generate distinct topic titles

Benefits of LLM Summarization

Efficiency	LLMs excel at swiftly processing and summarizing vast amounts of information, outperforming traditional manual methods.
Objectivity	LLM-based summarization provides a dynamic and objective approach, minimizing the influence of human perspectives and cognitive limitations.
Scalability	LLMs handle large volumes of data and easily scale for summarizing long documents or multiple sources of information.

Benefits of LLM Summarization

Accuracy	Achieving consistently increasing accuracy, LLMs are trained on massive datasets to deliver precise summaries.
Low-latency responses	Leveraging modern technology, LLMs generate rapid, low-latency responses, meeting real-time summarization needs.
Versatility	Applying their capabilities across various domains such as business, legal, and academia, LLMs offer a versatile solution for summarizing different types of content.

Limitations of Summarization

Quality of summaries

LLMs may not consistently generate accurate or unique insights, which can be crucial in some applications.

Complexity of datasets

LLMs can face challenges in training and deploying applications, especially when working with large and complex datasets.

Training costs

LLMs may incur prohibitive training costs, posing a challenge for smaller organizations or those with limited resources looking to adopt the technology.

Response latency

LLMs may exhibit high response latency, a concern for real-time applications such as summarizing live events or providing customer support.

Limitations of Summarization

Context limit

LLMs may encounter limitations in understanding and retaining context during summarization, resulting in a loss of information in the final summary.

Prompt fragility

LLMs can be sensitive to the choice of prompts, presenting challenges in developing reliable and consistent summarization solutions.

Repeatability

LLM summarization results can be challenging due to the inherent stochastic nature of the models and the choice of hyperparameters.

Summarization: Industrial Applications

Finance: LLMs summarize market reports and data.

Law: LLMs summarize legal documents and precedents.

Media and entertainment: LLMs summarize articles.

Marketing and advertising: LLMs summarize and generate marketing content.

Healthcare and enterprises: LLMs provide executive-level summaries of various functions.

Demo: Text Summarizer



Duration: 25 minutes

Overview:

In this demo, you will create an Arxiv paper summarizer that downloads a paper as a PDF, reads it in one shot, and generates a summary. You can use **The Impact of Generative Artificial Intelligence** paper as an example for demonstration.

Note

Please download the solution document from the Reference Material Section and follow the Jupyter Notebook for step-by-step execution.

DEMONSTRATION

Quick Check



How does the **refine** method work in LLM summarization?

A. Break-Summarize-Concatenate

B. Combine-String-Summarize

C. Segment-Refine-Map

D. Summarize-Combine-Refine

Content Translation

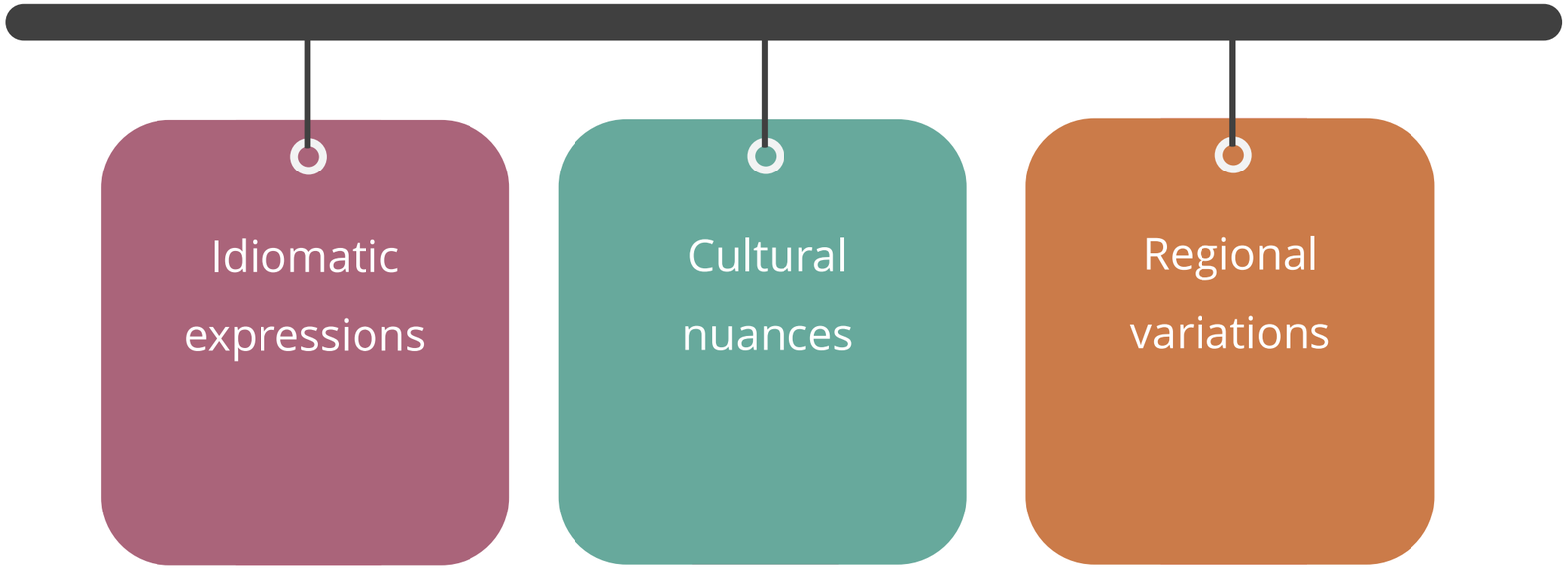
Content Translation

LLMs are increasingly used for content translation, offering several advantages and capabilities.



Content Translation

LLMs possess improved context awareness, allowing for a better understanding of:



Benefits of Content Translation

Multilingual support

LLMs can work with multiple languages to foster global communication and information access.

Improved context awareness

LLMs have a strong ability to understand and generate text based on context, resulting in more accurate and culturally appropriate translations.

Reduced bias

LLMs provide an opportunity to address biases that may exist in the model's default settings, which leads to more fair and unbiased translations.

Efficiency

LLMs offer efficient and cost-effective language translation services to reduce the need for manual intervention and speed up processes.

Benefits of Content Translation

Consistency and quality

LLMs offer consistency and quality in various tasks, including content creation, quality assurance, and data analysis.

Accessibility and inclusivity

LLMs can translate content into multiple languages, making information accessible to a global audience.

Personalized assistance

LLMs have the potential to offer personalized assistance and recommendations to enhance the user experience.

Limitations of Content Translation

Accuracy and quality

LLMs are not perfect and can occasionally produce incorrect translations.

Biases

LLMs may have biases in their default settings, which can lead to unintentional biases in translations.

Legal and regulatory compliance

LLM-based language translation presents challenges when working with sensitive or regulated content.

Context awareness

LLMs may struggle with understanding and retaining context during translation.

Content Translation: Industrial Applications

Global businesses: LLMs are transforming the way documents are translated for global businesses and overcoming language barriers.

E-commerce: LLMs are highly valuable for e-commerce platforms in translating product descriptions and customer reviews automatically.

Marketing and advertising: LLMs create multilingual content, such as dynamic user notifications, localized marketing materials, or personalized messages.

Travel and tourism: LLMs provide real-time, high-quality translations that are often on par with those produced by human translators.

Language service industry: LLMs improve the efficiency and quality of machine translation, reducing the need for manual intervention and speeding up processes.

Quick Check



Which of the following capabilities of Large Language Models (LLMs) is enhanced due to their improved context awareness in content translation?

A. Lexical disambiguation

B. Syntactic parsing

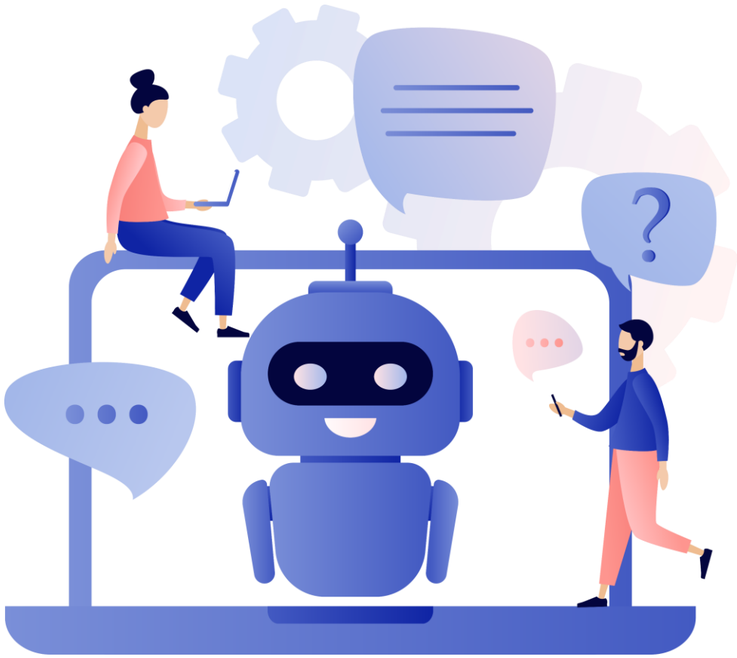
C. Idiomatic expressions

D. Semantic role labelling

Chatbots and Virtual Assistants

Chatbots and Virtual Assistants

LLMs have enhanced conversational AI systems, allowing chatbots and virtual assistants to engage in more natural, context-aware, and meaningful conversations with users.



Benefits of Chatbots and Virtual Assistants

Cost savings

LLM-powered chatbots and virtual assistants can handle simple service requests to reduce the need for humans and lower costs.

Improved user engagement

LLM enables chatbots and virtual assistants to provide personalized, interactive experiences that engage users and keep them coming back.

Increased accessibility

LLM can make chatbots and virtual assistants more accessible to people with disabilities, such as those who are visually impaired or have limited mobility.

Real-time assistance

LLM-powered chatbots and virtual assistants can provide real-time assistance to users to help them resolve issues quickly and efficiently.

Benefits of Chatbots and Virtual Assistants

Multilingual support

LLM enables chatbots and virtual assistants to understand and respond to requests from global customers in multiple languages.

Enhanced customer support

LLM-powered chatbots are handling routine inquiries, troubleshooting issues, and providing relevant information with remarkable accuracy.

Personalized assistance

LLM can make chatbots and virtual assistants more accessible to people with disabilities, such as those who are visually impaired or have limited mobility.

24/7 availability

LLM-powered chatbots can operate round the clock, providing support and information at any time to improve customer service and satisfaction.

Chatbots and Virtual Assistants: Industrial Applications

Customer support: LLMs enhance customer service with 24/7 support, personalized recommendations, and reduced response times.

E-commerce: LLM-powered chatbots streamline purchases, improve efficiency, and enhance the overall customer experience.

Healthcare: LLM-powered virtual assistants automate tasks, handle inquiries, and offer real-time support.

Retail: LLM-powered chatbots are used for natural language conversations, understanding queries, and improving customer support.

Finance: LLM-powered chatbots are used for personalized trading, efficient onboarding, market predictions, and smart wealth management.

Demo: MultiPDF QA Retriever with ChromaDB and LangChain



Duration: 30 minutes

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Overview:

In this demo, you will learn how to use LangChain to create a MultiPDF retriever with ChromaDB. This demo is performed on new generative AI research paper PDFs. You will understand how to load and process documents, create a database, make a retriever, create a chain, and use the retriever to ask questions and get answers.

Note

Please download the solution document from the Reference Material Section and follow the Jupyter Notebook for step-by-step execution.

Quick Check



Which of the following benefits are facilitated by Large Language Models (LLMs) in conversational AI systems?

A. Semantic mapping

B. Lexical disambiguation

C. Syntactic parsing

D. Personalized assistance

Sentiment Analysis

Sentiment Analysis

LLMs can understand the context in which words and phrases are used and perform sentiment analysis in real time.



Benefits of Sentiment Analysis

Improved decision-making	Companies can process vast amounts of data, enabling them to track brand sentiment and make data-driven decisions.
Personalized services	Companies can enhance their understanding of customer sentiment and personalize their services accordingly.
Improved customer support	Companies can respond promptly to customer needs, identify potential pain points, and provide more empathetic support.

Limitations of Sentiment Analysis

Cultural nuances	LLMs may struggle to capture cultural and contextual nuances that influence sentiment analysis, leading to incorrect sentiment predictions.
Sentiment ambiguity	LLMs may struggle with the inherent ambiguity in human language, leading to the potential misinterpretation of typos, synonyms, abbreviations, and mixed sentiments.
Lack of individual variation	LLMs can struggle with capturing individual variations in sentiment expression, leading to less accurate sentiment analysis.

Demo: Sentiment Analysis



Duration: 30 minutes

Overview:

In this demo, we aim to utilize two different tools to analyze the sentiment of a given text.
The two tools are:

- OpenAI's GPT-3 API
- TextBlob

Note

Please download the solution document from the Reference Material Section and follow the Jupyter Notebook for step-by-step execution.

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Introduction to Benchmarking

What Is Benchmarking?

Benchmarking in large language models (LLMs) is a process of evaluating and comparing the performance of various LLMs through standardized tests or tasks.



LLM Benchmarking: Steps

Benchmarking in LLMs involves the following steps:

Step 01

Benchmark selection

Step 02

Dataset preparation

Step 03

Model training and fine-tuning

Step 04

Model evaluation

Step 05

Comparative analysis

LLM Benchmarking: Steps

Benchmarking in LLMs involves the following steps:

Step 01

Benchmark selection

A benchmark is chosen to encompass a broad spectrum of challenges related to language.

Step 02

Dataset preparation

Specific datasets are meticulously curated for each benchmark task, comprising training, validating, and testing sets.

LLM Benchmarking: Steps

Benchmarking in LLMs involves the following steps:

Step 03

Model training and fine-tuning

The LLMs undergo training and fine-tuning using the meticulously prepared datasets.

Step 04

Model evaluation

The performance of the LLMs is assessed against the benchmark tasks.

LLM Benchmarking: Steps

Benchmarking in LLMs involves the following steps:

Step 05 Comparative analysis	The outcomes are scrutinized to compare the effectiveness of distinct LLMs.

Note

Benchmarking aims to thoroughly measure and compare LLMs, highlighting their strengths and weaknesses.

Quick Check



Which step assesses model performance against chosen benchmarks in language model development?

A. Benchmark selection

B. Dataset preparation

C. Model training and fine-tuning

D. Model evaluation

Benchmarks for Evaluating LLMs

Benchmarks for Language Models

Metrics such as ROUGE and BLEU scores are used to evaluate language models for summarization or text-generation tasks.



Demo: ROUGE Benchmark



Duration: 40 minutes

Overview:

This demo is designed to read a PDF file and a summary of that file, and then compute the ROUGE scores for the summary by comparing it with the original document. The ROUGE scores provide a measure of the quality of the summary.

Note

Please download the solution document from the Reference Material Section and follow the Jupyter Notebook for step-by-step execution.

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Need for New Benchmarks

LLMs transcend traditional language models, showcasing unparalleled proficiency in tasks beyond linguistic understanding.



- Existing benchmarks for language models are inadequate for evaluating these advanced LLMs.
- A compelling need exists for the creation of new benchmarks specifically tailored to assess the enhanced functionalities of LLMs.

Benchmarks for LLMs

There are some metrics to measure these models with billions of parameters:



Holistic Evaluation of Language Models (HELM)

It was developed by the Center of Research on Foundation Models (CRFM) at Stanford University.

The HELM benchmark tasks include:



General Language Understanding Evaluation (GLUE)

It is a collection of nine different natural language understanding tasks, introduced by researchers at Google AI.

The GLUE benchmark tasks include:

- Natural language interface
- Sentiment analysis
- Corpus of linguistic acceptability
- Sentence similarity
- Textual entailment
- Multi-genre natural language interface
- Paraphrasing
- WordNet hypernym
- Question natural language interface

GLUE Benchmark Tasks

Natural language interface (NLI)	This task involves determining whether a given hypothesis matches, opposes, or doesn't relate to a given statement.
Sentence similarity (STS)	This task involves assessing the degree of similarity in meaning between two given sentences.
Paraphrasing (QQP)	This task involves establishing whether two sentences serve as paraphrases of one another.

GLUE Benchmark Tasks

Sentiment analysis (SST-2)	This task involves determining if a given sentence carries a positive, negative, or neutral sentiment.
Textual entailment (RTE)	This task involves determining if a provided hypothesis is implied by a given premise.
WordNet hypernym (WNLI)	This task involves determining if a given word is a broader or more general term (hypernym) in comparison to another provided word.

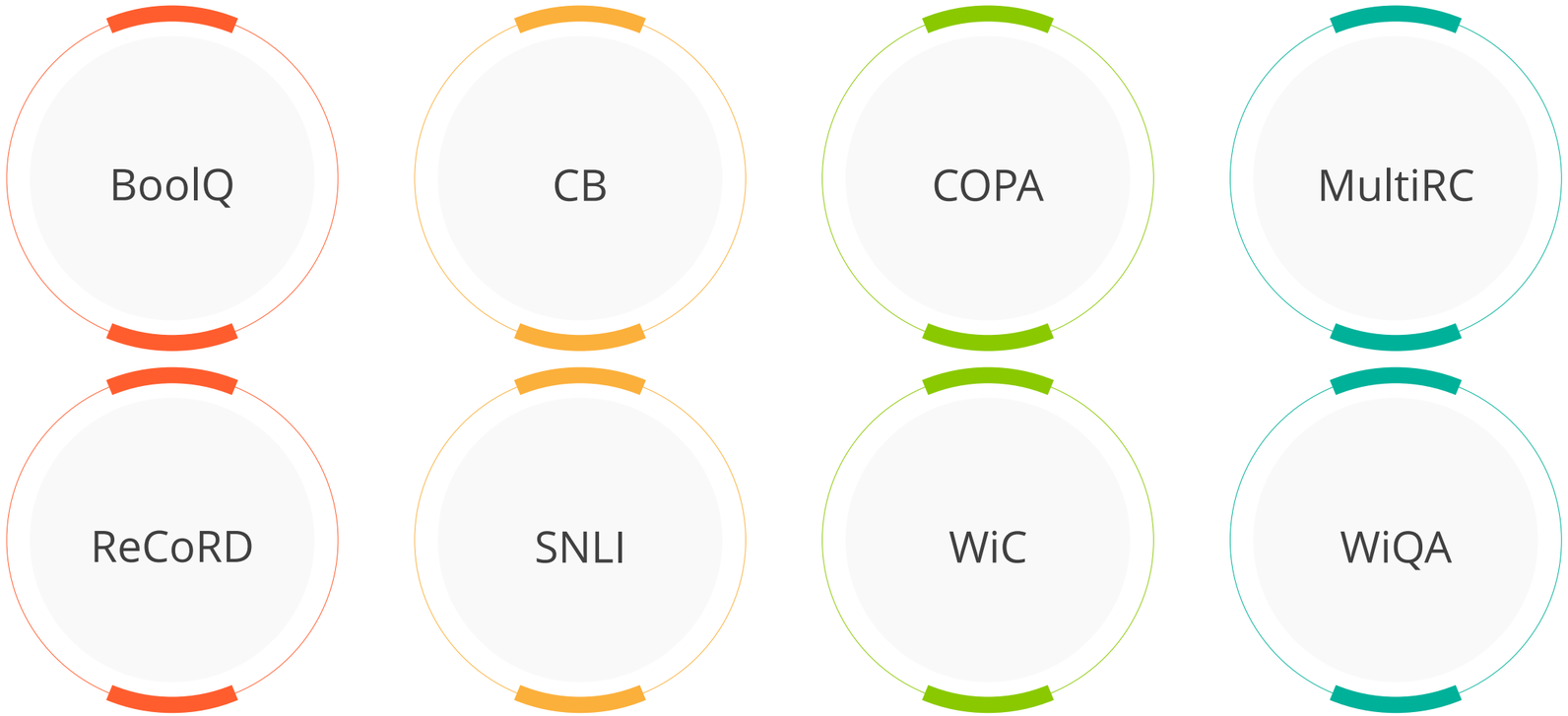
GLUE Benchmark Tasks

Corpus of linguistic acceptability (CoLA)	This task involves determining the grammatical correctness of a provided sentence.
Multi-genre natural language interface (MNLI)	This task is like NLI, but the hypothesis might consist of multiple sentences.
Question natural language interface (QNLI)	This task is like NLI, but the hypothesis is consistently framed as a question.

SuperGLUE

It is a collection of eight different natural language understanding tasks introduced by researchers at Facebook AI, Google AI, the University of Washington, and New York University.

The SuperGLUE benchmark tasks include:



SuperGLUE Benchmark Tasks

BoolQ

This task involves responding to factual yes-or-no questions.

CB

This task involves determining whether a given sentence is common sense or not.

COPA

This task involves deducing an accurate response to a question from the context provided in two sentences.

MultiRC

This task involves responding to multiple-choice questions related to a series of reading comprehension passages.

SuperGLUE Benchmark Tasks

ReCoRD

This task involves completing the blanks in a brief passage of text.

SNLI

This task involves determining if a provided sentence is entailed, contradictory, or neutral to a given premise.

WiC

This task involves selecting the most appropriate word or phrase to complete a sentence with a blank.

WiQA

This task involves responding to questions related to articles on Wikipedia.

Beyond the Imitation Game Benchmark (BIG-bench)

It serves as a collaborative benchmark designed to examine large language models and anticipate their future capabilities.

The BIG-bench benchmark tasks include:

- Text generation
- Translation
- Question answering
- Code generation
- Commonsense reasoning
- Social reasoning
- Logical reasoning

Quick Check



Which SuperGLUE task entails responding to multiple-choice questions?

A. COPA

B. MultiRC

C. BoolQ

D. Record

Guided Practice



Overview

Duration: 45 minutes

In this activity, you will create a PDF summarizer and will benchmark the summary using ROUGE. This guided practice aims to reinforce your skills and familiarity with these technologies, ensuring a more comprehensive understanding of future practical applications in the respective field.

GUIDED PRACTICE

Key Takeaways

- LLMs enhance conversational AI, making chatbots and virtual assistants more natural and meaningful by understanding context and performing real-time sentiment analysis.
- LLM summarization controls tone and style, obtains coherent topics, and generates distinct titles, which is useful for content creation.
- LLMs improve content translation, making it more efficient and accurate across languages.
- The HELM, GLUE, SuperGLUE, and BIG-bench benchmarks comprehensively evaluate LLM performance and capabilities.



Q&A

